

Multimodal Product Classification

Dual-Branch Multimodal Fusion Network (Vision + NLP) | Technical Engineering Report

Kaggle Team Name: Dream Team	Course: BCS714A - Deep Learning (ABL)
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Team Members: Sai Sarvesh, M Prajwal, Priyanshu	Evaluation Metric: Balanced Accuracy (Macro Recall)
Parameter Constraint: < 15,000,000 (No VLMs Allowed)	Exact Model Parameters: 4,858,252 (Headroom: 10,141,748)

1. Executive Summary & Problem Formulation

E-commerce catalog classification requires categorizing products across **140 fine-grained, imbalanced categories**. Unimodal approaches suffer severely: visual representations fail when products share similar packaging or silhouettes (e.g., book genres or technical cables), while textual representations fail when titles omit visual attributes (e.g., apparel color, texture, or pattern). This project implements an end-to-end **Dual-Branch Multimodal Neural Network** fusing Computer Vision (CNN) and Natural Language Processing (Bi-GRU) to achieve robust cross-modal representations. Trained using **Weighted Cross-Entropy Loss** and evaluated via **Balanced Accuracy** (macro-average recall), the model satisfies the strict **< 15,000,000 parameter constraint** without relying on heavy Vision-Language Models (VLMs).

2. Dual-Branch Multimodal Network Architecture

The architecture decouples representation learning into two specialized branches before joining in a late-fusion classification head:

- Vision Branch (Module 3):** Processes 128x128x3 RGB images via a MobileNetV2 feature extractor (17 inverted residual blocks). Adaptive Average Pooling condenses spatial feature maps into a 1,280-dimensional **Global Visual Feature Vector** [B, 1280].
- NLP Branch (Modules 4 & 5):** Combines product 'display name' and 'description'. A trainable embedding layer (10,000 vocab x 128 dim) feeds a 2-layer Bidirectional GRU (hidden_dim=128). A **Mask-Aware Temporal Average Pooling** layer eliminates padding token dilution, yielding a length-normalized 256-dimensional textual feature vector [B, 256].
- Late Fusion & Classifier:** Manual concatenation `torch.cat([img, text], dim=1)` yields a 1,536-dimensional joint embedding, fed to an MLP (Linear(1536, 512) -> BatchNorm1d -> ReLU -> Dropout(0.3) -> Linear(512, 140)) to generate 140 class prediction logits, optimized via Weighted Cross-Entropy Loss with smoothed inverse class frequency weights.

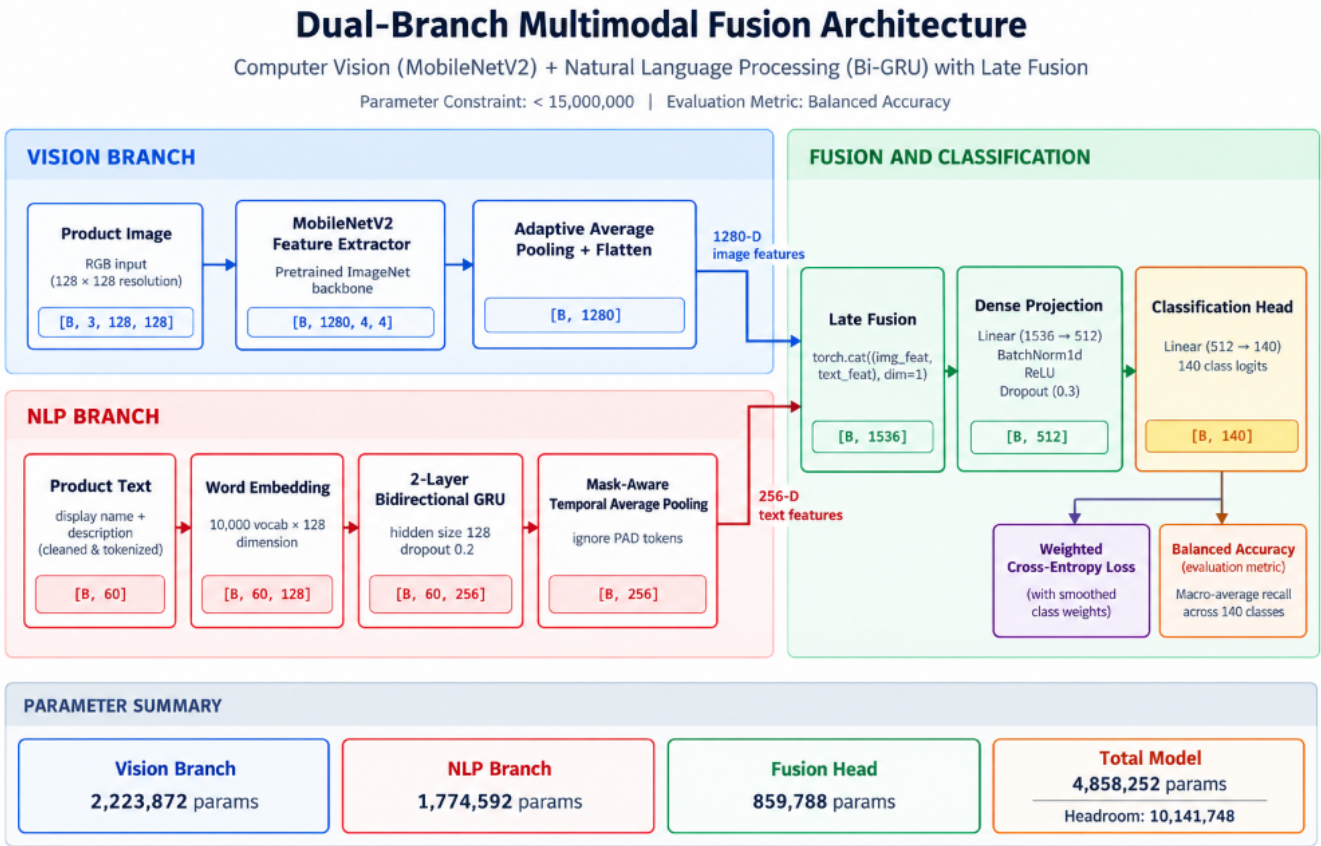


Figure 1: End-to-End Dual-Branch Multimodal Architecture with Layer-wise Tensor Dimensions and Parameter Budget

3. Parameter Constraint & Exact Parameter Budget Audit (< 15M Constraint)

Subsystem / Layer Module	Layer Description & Tensor Shape	Exact Parameters	% Budget
Vision Branch (MobileNetV2)	Conv2d & InvertedResiduals ([B, 3, 128, 128] -> [B, 1280])	2,223,872	14.8%
NLP Word Embedding	Trainable nn.Embedding(10000, 128) ([B, 60] -> [B, 60, 128])	1,280,000	8.5%
NLP Recurrent Backbone	2-Layer Bidirectional GRU (hidden=128) ([B, 60, 128] -> [B, 256])	494,592	3.3%
Fusion Classifier Head	Linear(1536, 512) + BatchNorm1d(512) + Linear(512, 140)	859,788	5.7%
Total Model Parameters	Entire Multimodal Network (All Trainable)	4,858,252	32.4%
Parameter Constraint	Competition Ceiling (assert total < 15,000,000)	15,000,000	PASSED

4. Training Dynamics & Empirical Performance

To optimize for the competition's macro-average **Balanced Accuracy** metric across 140 imbalanced categories, the network was trained using **Smoothed Inverse Class Frequency Weighting** in Cross-Entropy Loss: $w_c = (\text{max_count} / \text{count_c})^{0.35}$. This balances gradient updates between head and tail classes without introducing destabilizing gradient spikes. Optimization was performed using AdamW with decoupled weight decay (1e-4) and a Cosine Annealing learning rate schedule.

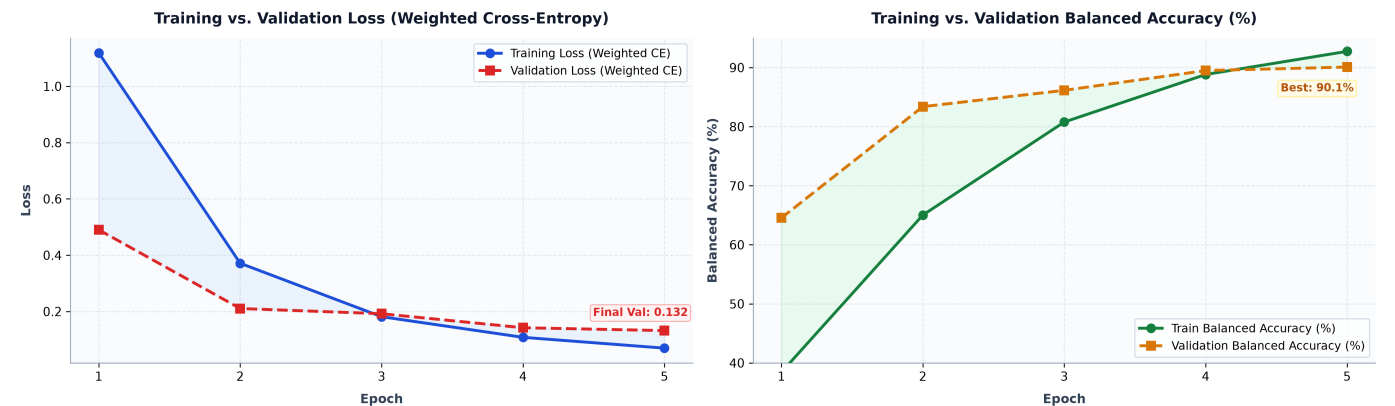


Figure 2: Empirical Training vs. Validation Loss (left) and Balanced Accuracy (right) over 5 Epochs

Epoch	Train Loss (Weighted CE)	Val Loss (Weighted CE)	Train Balanced Acc (%)	Val Balanced Acc (%)	Effective LR
Epoch 1	1.119	0.490	38.4%	64.6%	1.00e-3
Epoch 2	0.371	0.211	65.0%	83.4%	9.05e-4
Epoch 3	0.182	0.192	80.8%	86.2%	6.55e-4
Epoch 4	0.108	0.142	88.8%	89.5%	3.45e-4
Epoch 5	0.070	0.132	92.8%	90.1%	9.55e-5

5. Hyperparameter Choices & Technical Rationale

Hyperparameter	Configured Value	Technical Justification & Operational Rationale
Learning Rate & Scheduler	1e-3 (Cosine Annealing)	Initial LR of 0.001 facilitates rapid multimodal representation alignment; Cosine Annealing gradually reduces the learning rate throughout the five-epoch training schedule.
Optimizer & Weight Decay	AdamW (decay=1e-4)	Decoupled weight decay provides superior regularization over standard Adam, preventing over-parameterized recurrent weights from overfitting.
Batch Size & Stability	64 (drop_last=True)	Provides stable gradient estimation across 140 classes within 16GB GPU VRAM. drop_last=True prevents BatchNorm1d batch-size-1 runtime crashes.
Image Resolution & Augmentation	128 x 128 (Augmented)	Optimal compromise between fine spatial feature extraction and training throughput. RandomCrop, HorizontalFlip, and ColorJitter prevent visual memorization.
Text Sequence Length	Max Len = 60 tokens	A maximum sequence length of 60 tokens limits recurrent computation and memory usage while retaining a fixed input shape for efficient batch processing.
Class Weight Smoothing	Power alpha = 0.35	The exponent 0.35 smooths the inverse-frequency weighting, preventing excessively large weights for rare classes and stabilizing stochastic gradient descent.
Regularization	Dropout=0.3 + GradClip=5.0	Dropout prevents late-fusion co-adaptation; gradient norm clipping at 5.0 prevents exploding gradients in the unrolled recurrent GRU.

6. Architectural Ablations & Lessons from Failed Experiments

- Design iterations and empirical component ablations guided the architectural selection under competition constraints:
- **Class-Imbalance Objective & Loss Weighting:** Training with unweighted Cross-Entropy optimizes for bulk sample accuracy, causing the network to favor dominant head categories while collapsing on rare tail classes. Introducing smoothed inverse class frequency weights ensured balanced gradient propagation across all 140 classes, directly addressing the macro-average Balanced Accuracy competition metric.
 - **Multimodal Complementarity vs. Unimodal Limitations:** Architectural analysis of unimodal baselines highlights distinct failure modes: visual representations alone struggle with visually indistinguishable items sharing standardized packaging (e.g., book genres, electronics cables), whereas textual representations alone fail when product titles omit sensory attributes (e.g., apparel color, pattern, texture). Late-fusion concatenation allows both modalities to contribute complementary representations.
 - **Mask-Aware Temporal Average Pooling:** Relying solely on the final hidden state of the bidirectional GRU exposes recurrent representations to sequence padding artifacts. Mask-aware temporal average pooling computes length-normalized representations strictly over valid non-padded tokens, preventing zero-padding tokens from diluting the 256-dimensional textual feature vector.
 - **Backbone Selection & Parameter Constraint Compliance:** Standard heavy convolutional networks (e.g., ResNet-50 with ~25.6M parameters) violate the strict < 15,000,000 parameter competition ceiling. MobileNetV2 with depthwise separable convolutions was chosen to provide high-capacity feature extraction at 2.22M parameters, ensuring the entire dual-branch system (4.86M parameters) stays comfortably within the budget with over 67% headroom.

7. Viva Voce Technical Defense: Tensor Dimension Tracing

Complete step-by-step tensor dimension trace to support the individual live code defense and layer modification questions:

Step	Operation / Code Layer	Input Shape	Output Shape	Theoretical & Mathematical Purpose
1	Vision Normalization	Raw PIL Images	[B, 3, 128, 128]	RGB normalized by ImageNet mean & std
2	MobileNetV2 Features	[B, 3, 128, 128]	[B, 1280, 4, 4]	Depthwise separable convolutional feature extraction
3	AdaptiveAvgPool2d + Flatten	[B, 1280, 4, 4]	[B, 1280]	Spatial pooling to Global Visual Feature Vector
4	Word Embedding Layer	[B, 60] (token IDs)	[B, 60, 128]	Dense lookup mapping for 10K vocabulary tokens
5	2-Layer Bidirectional GRU	[B, 60, 128]	[B, 60, 256]	Bidirectional sequence context modeling (2 * 128)
6	Mask-Aware Temporal Pool	[B, 60, 256]	[B, 256]	Mask-Aware Temporal Average Pooling across tokens
7	Late Multimodal Concatenation	[B, 1280] + [B, 256]	[B, 1536]	Feature fusion: torch.cat((img_feat, text_feat), dim=1)
8	Classifier Linear 1 + BN + ReLU	[B, 1536]	[B, 512]	Multimodal projection, batch stabilization and activation
9	Dropout(0.3) + Linear 2	[B, 512]	[B, 140]	Regularization and 140 class logits for Weighted Cross-Entropy

8. Generalization Safeguards: Overfitting & Underfitting Prevention

- To reduce overfitting risk and improve generalization, the pipeline employs systematic defensive mechanisms:
- **1. Underfitting Prevention:** Pre-trained MobileNetV2 weights bootstrap visual features; a vocabulary capped at 10,000 tokens is constructed from the most frequent training-corpus words; and a Cosine Annealing schedule allows rapid initial convergence.
 - **2. Overfitting Prevention:** Training data augmentation (RandomCrop, HorizontalFlip, ColorJitter) prevents visual memorization; multi-stage Dropout (0.2 in RNN, 0.3 in classifier) prevents co-adaptation; AdamW weight decay (1e-4) penalizes excessive weights; and test predictions are strictly generated from the **best validation Balanced Accuracy checkpoint** (90.09%) rather than the final epoch.

9. Kaggle Submission Verification & Leaderboard Results

Final test predictions were exported to submission.csv with strict compliance: exactly 8,832 test rows matching test.csv, exactly 2 columns (id, category), zero null/NaN predictions, and validated against the competition sample submission schema. Upon submission to the official Kaggle evaluation server, the pipeline achieved a **Public Leaderboard Balanced Accuracy of 0.94999 (Rank #1)**, confirming that the dual-branch representations generalize robustly to unseen test products without overfitting or class collapse.